

HARD & SOFT 2026 SUCEAVA

17.05.2026 – 24.05.2026

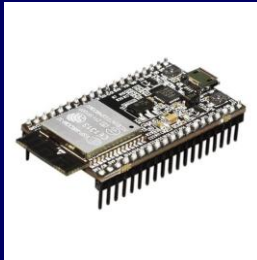
Cognizant Mobility Romania

Smart Fall Detection & Vital Monitoring Device for Elderly Care

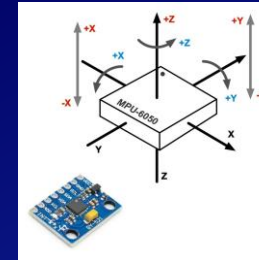
- Design and build a reliable embedded healthcare system capable of monitoring vital signs and detecting accidental falls in real time.
- The system must operate autonomously, process data locally (edge computing), and react reliably under uncertain real-world conditions.
- Participants are encouraged to combine embedded systems, intelligent decision-making, robustness engineering and security-aware design principles into a coherent standalone device.

WHAT YOU GET

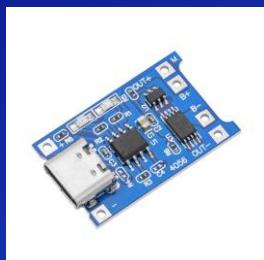
ESP32-DevKitC



Accelerometer +
Gyroscope Module



Battery & Charging
Module



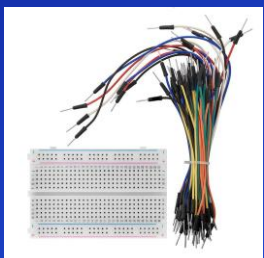
HR & SpO₂ Sensor



OLED Display



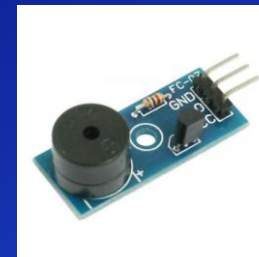
Breadboard Kit



Temperature Sensor



Buzzer & LEDs



Access to 3D printing facilities for
enclosure prototyping

WHAT YOU HAVE TO DO



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Key Objectives

- Monitor HR, SpO₂ and body temperature in real time
- Detect accidental falls using sensor fusion techniques
- Generate local alarms and event notifications
- Operate autonomously on battery power
- Minimize false alarms during normal activities
- Demonstrate reliable real-time behavior

Highlights

- Challenges your ability to build reliable embedded systems
- Encourages intelligent edge processing and explainable logic
- Security and robustness are considered part of the engineering solution
- Unknown live validation scenarios may be introduced during final demonstrations

THE JURY EXPECTS YOU TO

- ✓ Make purposeful, functional and technically justified design choices
- ✓ Use all essential hardware components provided in the kit and demonstrate coherent hardware–software integration.
- ✓ Build a truly autonomous system capable of operating without external cloud dependency.
- ✓ Pay attention to
 - Robustness
 - Reliability
 - Security
 - Power management
 - False-positive reduction
- ✓ Participants are encouraged to use AI tools responsibly, while maintaining full understanding of the implemented solution.

EVALUATION CRITERIA

Deliverables & Milestones

- Design Document: Submit a brief design PDF by Tuesday, 10:00 AM via email to the Jury.
- Jury Lab Visit: On Wednesday, the Jury will visit the lab for approximately 15 minutes to review work-in-progress.
- Final Report: Submit a final summary PDF by Friday, 8:00 AM via email to the Jury.
- Public Demonstration: On Friday morning, live final demonstration and technical evaluation

Evaluation Criteria

- Core functionality and reliability
- Fall detection performance
- Intelligent edge processing, security & robustness
- Hardware–software integration
- User experience and usability, technical presentation and documentation

JURY



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President:

Vlad VOICULESCU – Cognizant-Mobility GmbH, Munich

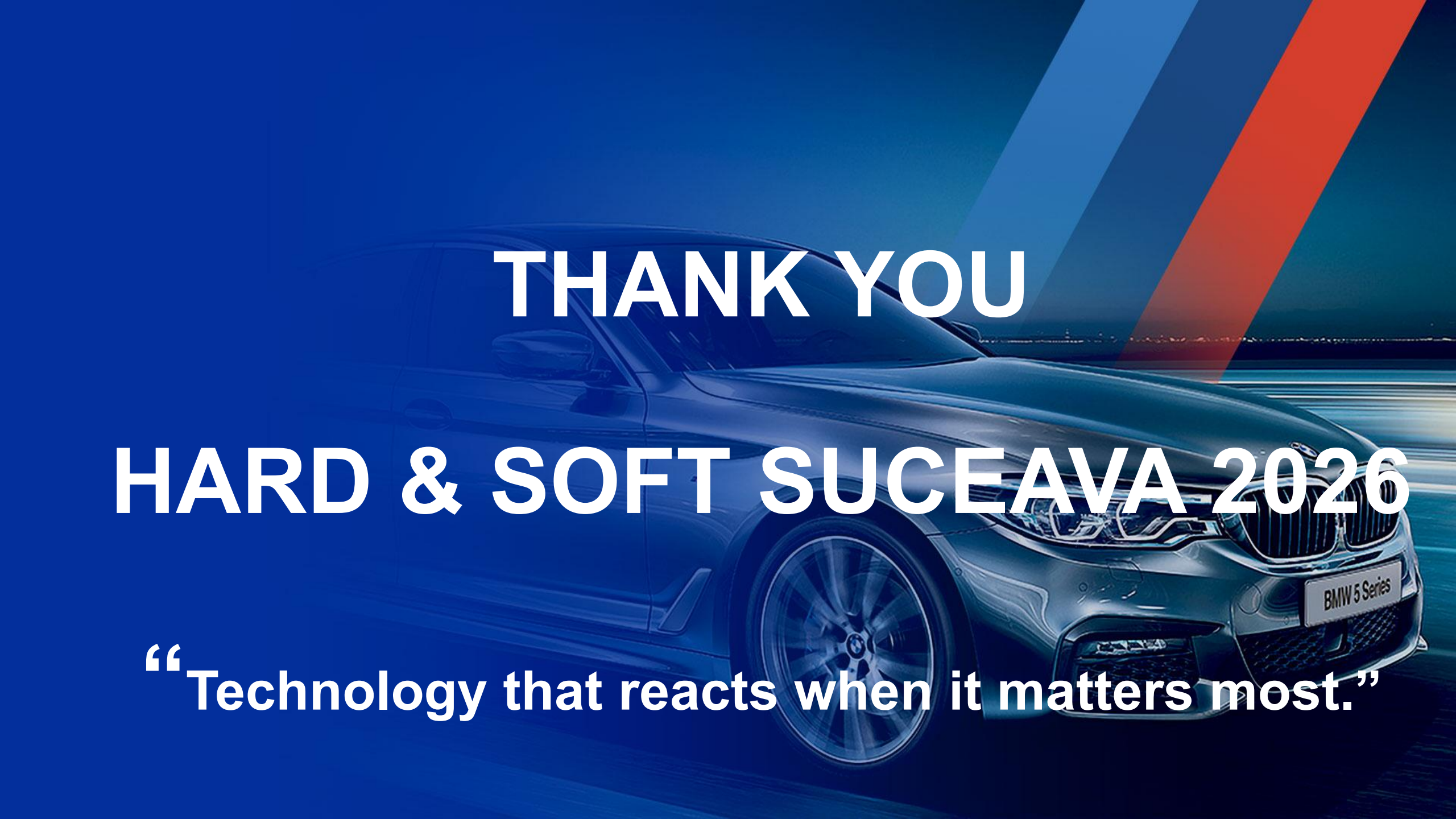
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THANK YOU

HARD & SOFT SUCEAVA 2026

“Technology that reacts when it matters most.”